Dominika Ďurovčíková, PhD

Cirriculum Vitae

Massachusetts Institute of Technology
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Research Positions

E. Margaret Burbidge Prize Postdoctoral Fellow — since Sep 2026
Department of Astronomy and Astrophysics, University of Chicago

Postdoctoral Research Specialist — Jun 2026 – Aug 2026
Cosmic Dawn Group, Massachusetts Institute of Technology
PI: Anna-Christina Eilers

Graduate Research Assistant — Sep 2022 – May 2026
Cosmic Dawn Group, Massachusetts Institute of Technology
PIs: Anna-Christina Eilers, Robert A. Simcoe

Graduate Research Assistant — Sep 2020 – Aug 2022
Quantum and Precision Measurements Group, Massachusetts Institute of Technology
PI: Vivishek Sudhir

Undergraduate Research Assistant — Sep 2017 – Jun 2020
Beecroft Institute of Particle Astrophysics and Cosmology, University of Oxford
PIs: Adrianne Slyz, Julien Devriendt

Undergraduate Intern — Jun 2019 – Sep 2019
Tearney Laboratory, Massachusetts General Hospital
PI: Guillermo J. Tearney

Laidlaw Scholar — Jun 2018 – Sep 2018
LIGO Laboratory, Massachusetts Institute of Technology
PI: Nergis Mavalvala

Undergraduate Intern — Jun 2017 – Aug 2017
Research Center for Quantum Information
PI: Daniel Nagaj

High-School Thesis — Mar 2015 – Jun 2016
M.R. Štefánik Observatory
PI: Karol Petrík

Education

PhD in Physics — 2020 - 2026
Massachusetts Institute of Technology, Cambridge, MA, USA
- Thesis: Black holes on the largest and the smallest scales
- Thesis advisors: Prof. Anna-Christina Eilers, Prof. Robert A. Simcoe
- Primary research area: high-redshift astrophysics
- Other research areas: quantum metrology (with Prof. Vivishek Sudhir)

Master of Physics (4-year MPhys) — 2016 - 2020
University of Oxford, Oxford, United Kingdom
- Degree classification: First class
- Thesis: Super-resolution galaxy images from generative adversarial networks
- Graduate concentration: theoretical physics, laser physics and quantum information processing

Awards & Honors

E. Margaret Burbidge Prize Postdoctoral Fellowship	2026
MIT School of Science Service Fellowship	2022
MIT Physics Graduate Service Award	2021
Bruno Rossi Graduate Fellowship	2020 - 2021
Scholarship of the College of the Blessed Mary of Winchester	2017 - 2020
Harvard-MIT Summer Institute for Biomedical Optics Certificate	2019
Institute of Leadership & Management (ILM) Certificate Level 3	2019
McKinsey&Company Next Generation Women Leaders Award	2019
Laidlaw Research and Leadership Scholarship	2018
Distinction in Physics	2017

Telescope Use

Keck/MOSFIRE (PI, 0.5 nights, $\$15k$) 2025
A potential weak-line quasar transition at $z \sim 6$

Magellan/FIRE,LDSS3 (PI, 1 nights) 2024
A candidate metal-poor absorption system at $z \sim 6$

Magellan/FIRE (PI, 7 nights) 2024
Exploring the connection between supermassive black hole lifetimes and the history of their galactic environment

Magellan/FIRE (PI, 1 night) 2023
Chronicling the reionization history with redshift $z \sim 6.5$ quasars

Co-I: JWST Cycle 2 GO #3079 (NIRSpec IFU), JWST Cycle 4 GO #6827 (NIRCam WFSS, MIRI), Magellan/LLAMAS, multiple Magellan/FIRE proposals

Observing experience: Magellan/FIRE (24 nights), Magellan/LDSS3 (0.5 night), Magellan/LLAMAS (2 nights), Keck/MOSFIRE (0.5 night)

Invited Talks

Konkoly Observatory & Eötvös Loránd University Group Meeting, Budapest, Hungary	06/2026
European Southern Observatory, Garching, Germany	01/2026
3KK (Three Kings' Conference), Bratislava, Slovakia	01/2026
KICP Group Meeting, University of Chicago, Chicago, IL	09/2025
ENIGMA Group Meeting, UC Santa Barbara, Santa Barbara, CA	05/2025
Astro Lunch, UC Santa Barbara, Santa Barbara, CA	05/2025
High-redshift Meeting, Harvard & Smithsonian Center for Astrophysics, Cambridge MA	04/2025
Slovak Technical University, Trnava, Slovakia	03/2025
IEEE Buenaventura Section	11/2024
Summer Conference on Particle Physics Solid State Physics, University of Tennessee, TN	07/2024
Science coffee at Charles University, Prague, Czech Republic	02/2024
State of the Universe seminar, Tata Institute of Fundamental Research, Mumbai, India	12/2023
Modern statistics of galaxies seminar, University Observatory of LMU, Munich, Germany	09/2023
MIT Kavli Institute Journal Club, Cambridge, MA	02/2022
Particle Physics/Astrophysics/Machine learning Seminar, Oxford, UK	02/2020

Conference Talks	i2i3: Still linking galaxy physics from ISM to IGM scales, Sesto, Italy	01/2026
	Highly Accreting Supermassive Black Holes Across all Cosmic Times, Santiago, Chile	12/2025
	The First Gigayear(s) Conference, Hilo, HI	10/2024
	EREBUS/JWST workshop, Hilo, HI	09/2024
	First Stars VII, New York City, NY	05/2024
	Reionization in the Summer, Heidelberg, Germany	06/2023
	* First Light Conference, Cambridge, MA	06/2023
	Conference on Lasers and Electro-Optics (CLEO), San Jose, CA	05/2022
	* MIT QSEC Annual Research Conference, Cambridge, MA	02/2022
	Global Young Scientists Summit, Singapore	01/2021
	Summer All Zoom Epoch of Reionization Astronomy Conference (SAZERAC)	07/2020
	* Royal Society-FAPESP Frontiers of Science Meeting, São Paulo, Brazil	03/2020
	* Wellman Scientific Retreat, Boston, MA	09/2019
	* Harvard-MIT Summer Institute for Biomedical Optics Poster Day, Boston, MA	08/2019
	Harvard-MIT Summer Institute for Biomedical Optics Presentations, Boston, MA	07/2019
	* Laidlaw Research and Leadership Programme Poster Event, Oxford, UK	10/2018
	MIT Kavli Institute Undergraduate Research Symposium Cambridge, MA	08/2018
	* poster presentations	

Conference Attendance	Quasar Bazaar Hackweek, Trieste, Italy	02/2026
	Boston-Area Black Hole Accretion Meeting, Harvard & Smithsonian Center for Astrophysics, Cambridge, MA	10/2023
	APS Virtual Division of Atomic, Molecular and Optical Physics (DAMOP) Meeting	06/2020
	First Light and Reionisation Epoch Meeting at Royal Astronomical Society, London, UK	02/2020
	FUTURE of Physics at California Institute of Technology, Pasadena, CA	11/2018

Public Talks	Slovak PRO Summit, Consulate General of Slovakia in New York, New York City, NY	09/2024
	Slovak Astrophysicists in Boston, Cambridge, MA	03/2024

Teaching & Service Work

Referee for: The Astrophysical Journal, Nature Astronomy, Physical Review Journals

Research Mentor | MIT Undergraduate Research Opportunities Program — 2023 - 2024
- Supervised two undergraduate students on a high-redshift quasar research project.

Co-director | MIT Astrogazers — 2023 - 2024
- Bringing telescopes to the streets of Cambridge and Boston (and beyond).

Local Organizing Committee Member | First Light Conference — 2023

Teaching Assistant | MIT Department of Mechanical Engineering — 2022
- Co-developed a new course on classical and quantum stochastic processes.
- Created and marked 7 problem sets, hosted weekly office hours, and marked final talks.

Vice-President for Admissions | MIT Physics Graduate Student Council — 2021 - 2022
- Oversaw and coordinated student initiatives related to improving admissions to the MIT Physics graduate program.
- Collaborated with the Physics Graduate Student Council leadership on improving the student experience at MIT Physics.

Student Leader | MIT Physics Graduate Admissions Advisory Council — 2020 - 2022
- Co-designed and launched three new student-led resources under the umbrella of PhysGAAP to improve the MIT Physics graduate admissions process.
- Collaborated with student leaders from other MIT departments to achieve a more uniform change in admissions across MIT.

Lecturer | Discover Summer Academy — 2020 - 2022
- Designed and taught a week-long course on quantum physics (twice) and on black holes (once) to high school students from Slovakia and Czech Republic.
- Facilitated team-building and reflection sessions in three teams of $\sim$10 students.

Co-Founder | EWAAB Nonprofit Organisation — 2019 - present
- Co-founded EWAAB as an initiative to support confidence in university-level women. We aim to encourage young women to step out of their comfort zone, to provide them with a set of leadership and communication skills to be able to do so, and to connect them to a global network of peers and supporters. Featured in the Scientific American and SME (the largest Slovak newspaper).
- Transformed the original initiative into a 501(c)3 nonprofit organization supported by a Board of Trustees and a team of volunteers.
- Co-designed the curriculum of the 2019/20 mentorship program and managed a successful launch of its inaugural year at 8 universities around the world, spanning Canada to Australia, together impacting 27 mentees in 6 countries in its first year.
- Impact to date: 280+ students across 15 institutions worldwide.

Publications

Full list can be accessed through NASA/ADS.

16. **Ďurovčíková**, Eilers, Ishikawa, Yue, Vestergaard, Davies, Schindler, Fan, Arrigoni Battaia, Volonteri, Simcoe, Hennawi, Blecha, Andika, Bosman, Bieri. BEES: Quasar lifetime measurements from extended rest-optical emission line nebulae at $z \sim 6$. ApJ 1003 17 (2026).
15. Schneider, Brunet, Gompertz, Turpin, Malesani, Godet, Levan, Daigne, Sarin, Rakotondrainibe, Martin-Carrillo, Palmerio, Thöne, Li, Saccardi, de Ugarte Postigo, Antier, Buat, **Ďurovčíková**, Izzo, Leung, Mo, Qiu, Vergani, Wang, Wei, Xin, Mochkovitch, Zhang, Cai, Campana, Coleiro, Cordier, D'Avanzo, Dagoneau, D'Elia, Dong, Götz, Guillot, Han, Hartmann, Huang, Huang, Jakobsson, Klotz, Lachaud, Lu, Maggi, De Pasquale, Piron, Salvaterra, Schanne, Sollerman, Tanvir, Vidadi, Wang, Wu, Xiong, Xu, Zafar, Zhang, Zhang, Zheng. XRF 241001A/SN 2024aiiq: A Faint Soft X-ray Transient Detected by SVOM with a Broad-Line Type Ic Supernova Revealed by JWST. arXiv:2604.20346 (2026).

14. Wolf, Bañados, Fan, Dumont, Davies, Rupke, Yang, Liu, Belladitta, Barth, Bosman, Costa, Davies, Decarli, **Ďurovčíková**, Eilers, Jun, Liu, Loiacono, Lupi, Mazzucchelli, Pudoka, Rojas-Ruiz, Schindler, Tee, Trakhtenbrot, Walter, Zhang. Shedding the envelope: JWST reveals a kiloparsec-scale [O III]-weak Balmer shell around a $z = 7.64$ quasar. A&A 707 A299 (2026).
13. Bosman, Álvarez-Márquez, Davies, Protušová, Hennawi, Yang, Spina, Colina, Fan, Östlin, Walter, Wang, Ward, Herrero, Barth, Belladitta, Boogaard, Caputi, Connor, **Ďurovčíková**, Eilers, Gómez, Hjorth, Jun, Langeroodi, Liu, Lupi, Mazzucchelli, Pye, Rinaldi, van der Werf, Volonteri. A close look at the black hole masses and hot dusty toruses of the first quasars with MIRI-MRS. arXiv: 2511.02902 (2025).
Submitted to The Astrophysical Journal.
12. **Ďurovčíková**, Eilers, Meyer, Farina, Bañados, Davies, Hennawi, Mazzucchelli, Simcoe, Walter. Quasar lifetime measurements from extended Lyα nebulae at $z \sim 6$. ApJ 990 174 (2025).
Featured in American Astronomical Society NOVA: "Are quasars growing in secret?"
11. **Ďurovčíková**, Eilers, Simcoe, Welsh, Meyer, Matthee, Ryan-Weber, Yue, Katz, Satyavolu, Becker, Davies, Farina. An extremely metal-poor Lyman-α emitter candidate at $z = 6$ revealed through absorption spectroscopy. ApJL 987 L33 (2025).
10. **Ďurovčíková**, Sudhir. Scheme for continuous force detection with a single electron at the level of 10^{-27} N. PR Applied 23(5) p.054088 (2025).
9. Greig, Bosman, Davies, **Ďurovčíková**, Fathivavsari, Liu, Meyer, Sun, D'Odorico, Gallerani, Mesinger, Ting. Blind QSO reconstruction challenge: exploring methods to reconstruct the Lyα emission line of QSOs. MNRAS 533(3) pp.3312–3343 (2024).
8. **Ďurovčíková**, Eilers, Chen, Satyavolu, Kulkarni, Simcoe, Keating, Haehnelt, Bañados. Chronicling the reionization history at $6 \lesssim z \lesssim 7$ with emergent quasar damping wings. ApJ 969 162 (2024).
Featured in MIT News: "When the lights turned on in the universe"
7. Soria, De, **Ďurovčíková**, Simcoe, Karambelkar, Hankins, Kasliwal, Sokoloski, Ashley, Babul, Lau, Moore, Ofek, Sharma, Soon, Travouillon. Magellan/FIRE spectroscopy of AT2023tow: Confirmation of a young, highly reddened Galactic Fe II nova with CO emission. ATel #16255 (2023).
6. Eilers, Simcoe, Yue, Mackenzie, Matthee, **Ďurovčíková**, Kashino, Bordoloi, Lilly. EIGER III. JWST/NIRCam observations of the ultra-luminous high-redshift quasar J0100+2802. ApJ 950 68 (2023).
5. Komori, **Ďurovčíková**, Sudhir. Quantum theory of feedback cooling of an anelastic macro-mechanical oscillator. PRA 105(4) p.043520 (2022).
4. Bosman, **Ďurovčíková**, Davies, Eilers. A comparison of quasar emission reconstruction techniques for $z \geq 5.0$ Lyman-α and Lyman-β transmission. MNRAS 503(2) pp.2077–2096 (2021).
3. Reiman, Tamanas, Prochaska, **Ďurovčíková**. Fully probabilistic quasar continua predictions near Lyman-α with conditional neural spline flows. arXiv: 2006.00615 (2020).
2. Katz, **Ďurovčíková**, Kimm, Rosdahl, Blaizot, Haehnelt, Devriendt, Slyz, Ellis, Laporte. New Methods for Identifying Lyman Continuum Leakers and Reionization-Epoch Analogues. MNRAS 498(1) pp.164–180 (2020).
1. **Ďurovčíková**, Katz, Bosman, Davies, Devriendt, Slyz. Reionization history constraints from neural network based predictions of high-redshift quasar continua. MNRAS 493(3) pp.4256–4275 (2020).